



JPK

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62530 PUTRAJAYA**

KERTAS PENERANGAN
(INFORMATION SHEET)

KOD DAN NAMA PROGRAM / <i>PROGRAM'S CODE AND NAME</i>	Z-009-2: 2015CORE ABILITIES	
TAHAP / LEVEL	2 (TWO)	
NO. DAN TAJUK KEBOLEHAN TERAS <i>ICORE ABILITY NO. AND TITLE</i>	CA04 HEALTH, SAFETY AND ENVIRONMENTAL AWARENESS	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	04.01 DEMONSTRATE HEALTH COMPLIANCE 04.02 DEMONSTRATE SAFETY COMPLIANCE 04.03 DEMONSTRATE ENVIRONMENTAL COMPLIANCE	
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TAJUK/TITLE: SAFE MACHINE OPERATION APPLICATION

TUJUAN/PURPOSE:

This Information Sheet is intended to describe on the ability to demonstrate level of understanding in Health, Safety and Environment adaptation and apply it to work activities. The person who is competent in this CA shall be able to adapt to HSE by executing his/her duties while observing Health, Safety and Environment Rules and Regulations.

PENERANGAN/INFORMATION:

1.0 Introduction To Training And Supervision Of Machine Operators

The employer or person in control of the workplace must not let anyone use a machine unless they have had training on:

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- i. the actual and potential hazards of the machine
- ii. all the precautions to be taken.

Unless the machine operator can operate the machine safely, then the employer or person in control should have the operator closely supervised by someone with the skills.

Employees must be supervised and trained by a competent person

When training people to use machinery, a trainer must explain:

- i. how to check and adjust the machine before starting it
- ii. how to stop and start the machine
- iii. how the machine works
- iv. what the machine does
- v. location and operation of other controls
- vi. actual and potential hazards and appropriate ways to control them
- vii. purpose of guards and other safety devices
- viii. correct use and adjustment of guards
- ix. correct work methods to be used
- x. how to recognise faults that could cause harm
- xi. limitations and capabilities of the machine
- xii. emergency procedures.

Take manufacturer's instructions into consideration when developing training programmes for operators.

Unless they are:

- i. in a public access area
- ii. under the direct supervision of an adult
- iii. on a guided tour of the area
- iv. in an area only used for the sale of goods or services.

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2.0 Use of Machinery – The Employee

When using machinery an employee is responsible for:

- i. their own health and safety
- ii. not harming others through their actions or inaction
- iii. following any safe work procedures their employer has in place
- iv. identifying and reporting hazards – and controlling them where possible
- v. using all guarding
- vi. keeping their work areas clear, clean and tidy
- vii. wearing and using all protective equipment and clothing
- viii. alerting their supervisor to any machinery faults or maintenance needs
- ix. telling their supervisor about any illness or condition that could stop or limit their ability to work with machinery – to keep others safe and help supervisors allocate work.

Machine operators should:

- i. check that the machinery they use is in sound working order
- ii. report immediately any problems to their supervisor
- iii. use any safety devices, guards, appliances, protective devices and any other methods used to make the machinery safe.

Where there is a risk of entanglement with machinery, people should:

- i. tie long hair back close to the head
- ii. not wear loose clothing.

3.0 Requirements for Safeguards

Safeguards must meet the following minimum general requirements to protect workers against mechanical hazards:

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Prevent contact. The safeguard must prevent hands, arms or any part of a worker's body or clothing from making contact with dangerous moving parts by eliminating the possibility of the operators or other workers placing parts of their bodies near hazardous moving parts.

Provide security. Workers should not be able to easily remove or tamper with the safeguard. Guards and safety devices should be made of durable material that will withstand the conditions of normal use and that are firmly secured to the machine.

Protect from falling objects. The safeguard should ensure that no objects can fall into moving parts and damage the equipment or become a projectile that could strike and injure someone.

Not create new hazards. A safeguard defeats its purpose if it creates a hazard of its own, such as a shear point, a jagged edge or an unfinished surface. The edges of guards, for example, should be rolled or bolted in such a way that they eliminate sharp edges.

Not create interference. Safeguards which impede workers from performing their jobs might soon be overridden or disregarded. If possible, workers should be able to lubricate machines without disengaging or removing safeguards. For example, locating oil reservoirs outside the guard, with a line leading to the lubrication point, will reduce the need to enter the hazardous area.

4.0 Safeguard Training

Even the most elaborate safeguarding system cannot offer effective protection unless workers know how to use it and why. Specific and detailed training is an important part of any effort to implement safeguarding against machine-related hazards. Proper safeguarding may improve productivity and enhance efficiency since it may relieve workers' apprehensions about injury. Safeguard training is necessary for new operators and maintenance or set-up personnel, when any new or altered safeguards are put in service, or when workers are assigned to a new machine or operation; it should involve instruction or hands-on training in the following:

- i. a description and identification of the hazards associated with particular machines and the specific safeguards against each hazard

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- ii. how the safeguards provide protection; how to use the safeguards and why
- iii. how and under what circumstances safeguards can be removed, and by whom (in most cases, repair or maintenance personnel only)
- iv. what to do (e.g., contact the supervisor) if a safeguard is damaged, missing or unable to provide adequate protection.

5.0 Methods Of Machine Safeguarding

There are many ways to safeguard machinery. The type of operation, the size or shape of stock, the method of handling, the physical layout of the work area, the type of material and production requirements or limitations will help to determine the appropriate safeguarding method for the individual machine. The machine designer or safety professional must choose the most effective and practical safeguard available.

Safeguards may be categorized under five general classifications: (1) guards, (2) devices, (3) separation, (4) operations and (5) other.

5.1 Safeguarding with guards.

There are four general types of guards (barriers which prevent access to danger areas), as follows:

Fixed guards. A fixed guard is a permanent part of the machine and is not dependent upon moving parts to perform its intended function. It may be constructed of sheet metal, screen, wire cloth, bars, plastic or any other material that is substantial enough to withstand whatever impact it may receive and to endure prolonged use. Fixed guards are usually preferable to all other types because of their relative simplicity and permanence (see **Table 1**).

Method	Safeguarding action	Advantages	Limitations
Fixed	Provides a barrier	- Suits many specific applications - In-plant construction is often possible - Provides maximum protection - Usually requires minimum maintenance - Suitable to high production, repetitive operations	- May interfere with visibility - Limited to specific operations - Machine adjustment and repair often require its removal, thereby necessitating other means of protection for maintenance personnel
Interlocked	Shuts off or disengages power and prevents starting of machine when guard is open; should require the machine to be stopped before the worker can reach into the danger area	- Provides maximum protection - Allows access to machine for removing jams without time-consuming removal of fixed guards	- Requires careful adjustment and maintenance - May be easy to disengage or bypass
Adjustable	Provides a barrier which may be adjusted to facilitate a variety of production operations	- Can be constructed to suit many specific applications - Can be adjusted to admit varying sizes of stock	- Operator may enter danger area: protection may not be complete at all times - May require frequent maintenance and/or adjustment - May be made ineffective by the operator - May interfere with visibility
Self-adjusting	Provides a barrier which moves according to the size of the stock entering danger area	Off-the-shelf guards are commercially available	- Does not always provide maximum protection - May interfere with visibility - May require frequent maintenance and adjustment

Table 1: Machine guards

In figure 1, a fixed guard on a power press completely encloses the point of operation. The stock is fed through the side of the guard into the die area, with the scrap stock exiting on the opposite side.

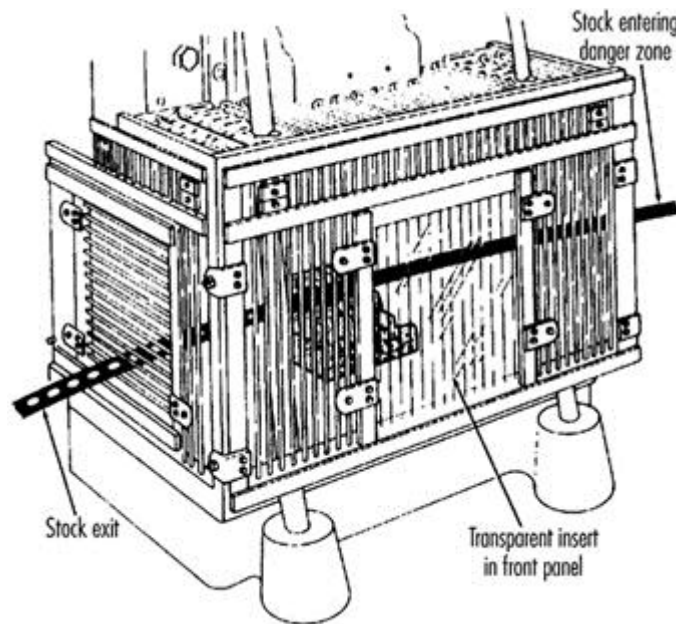


Figure 1. Fixed guard on power press

Figure 2 depicts a fixed enclosure guard which shields the belt and pulley of a power transmission unit. An inspection panel is provided on top to minimize the need for removing the guard.

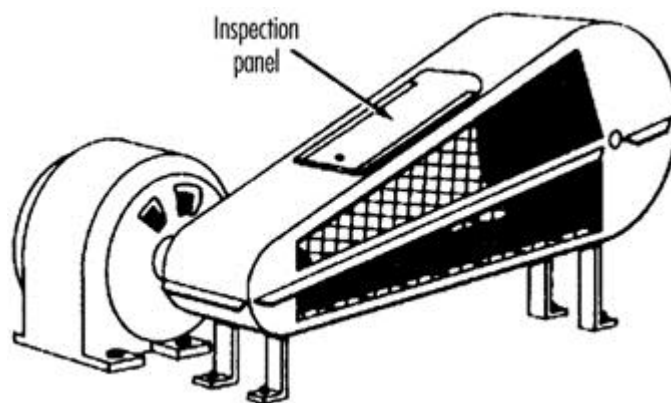


Figure 2. Fixed guard enclosing belts and pulleys

In figure 3, fixed enclosure guards are shown on a bandsaw. These guards protect operators from the turning wheels and moving saw blade. Normally, the only time the guards would be opened or removed would be for a blade change or for maintenance. It is very important that they be securely fastened while the saw is in use.

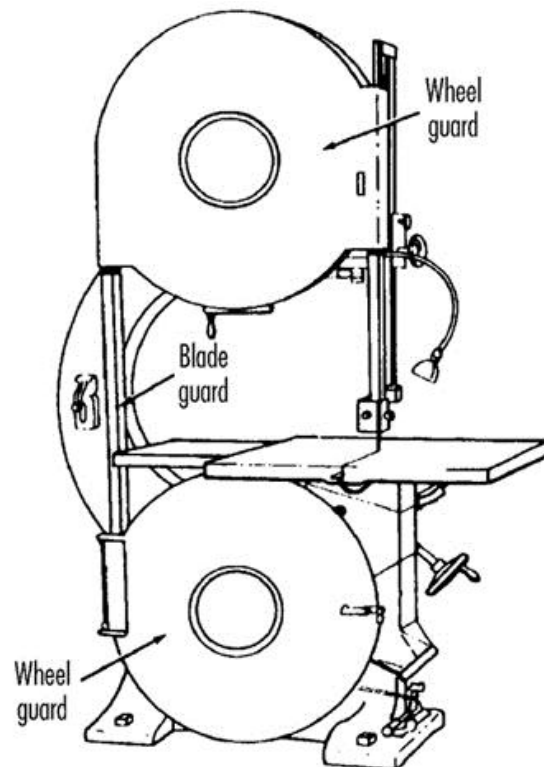


Figure 3. Fixed guards on band-saw

Interlocked guards. When interlocked guards are opened or removed, the tripping mechanism and/or power automatically shuts off or disengages, and the machine cannot cycle or be started until the interlock guard is back in place. However, replacing the interlock guard should not automatically restart the machine. Interlocked guards may use electrical, mechanical, hydraulic or pneumatic power, or any combination of these. Interlocks should not prevent “inching” (i.e., gradual progressive movements) by remote control, if required.

An example of an interlocking guard is shown in figure 4. In this figure, the beater mechanism of a picker machine (used in the textile industry) is covered by an interlocked barrier guard. This guard cannot be raised while the machine is running, nor can the machine be restarted with the guard in the raised position.

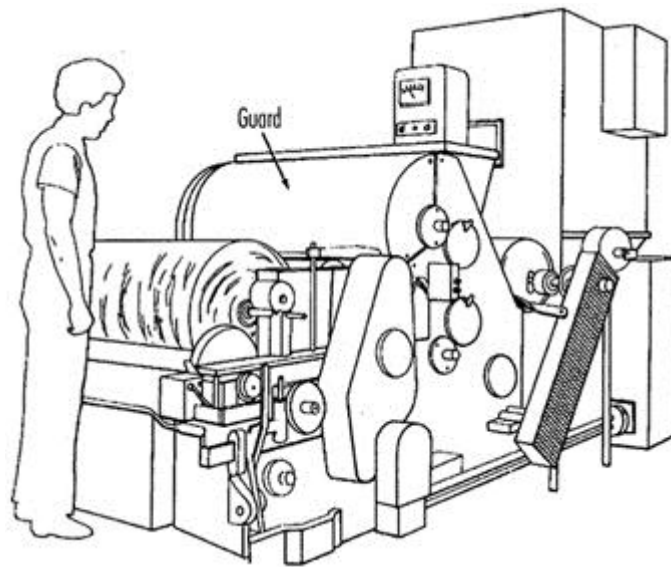


Figure 4. Interlocked guard on picker machine

Adjustable guards. Adjustable guards allow flexibility in accommodating various sizes of stock. Figure 5 shows an adjustable enclosure guard on a band-saw.

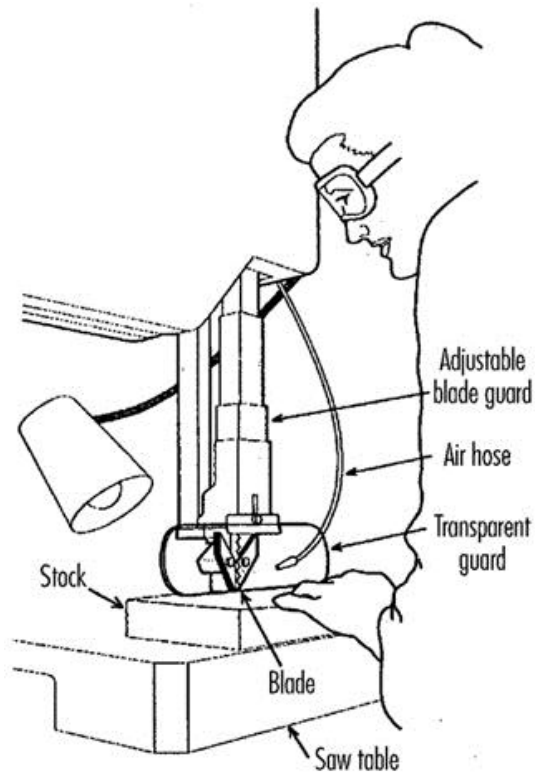


Figure 5. Adjustable guard on band-saw

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Self-adjusting guards. The openings of self-adjusting guards are determined by the movement of the stock. As the operator moves the stock into the danger area, the guard is pushed away, providing an opening which is large enough to admit only the stock. After the stock is removed, the guard returns to the rest position. This guard protects the operator by placing a barrier between the danger area and the operator. The guards may be constructed of plastic, metal or other substantial material. Self-adjusting guards offer different degrees of protection.

Figure 6 shows a radial-arm saw with a self-adjusting guard. As the blade is pulled across the stock, the guard moves up, staying in contact with the stock.

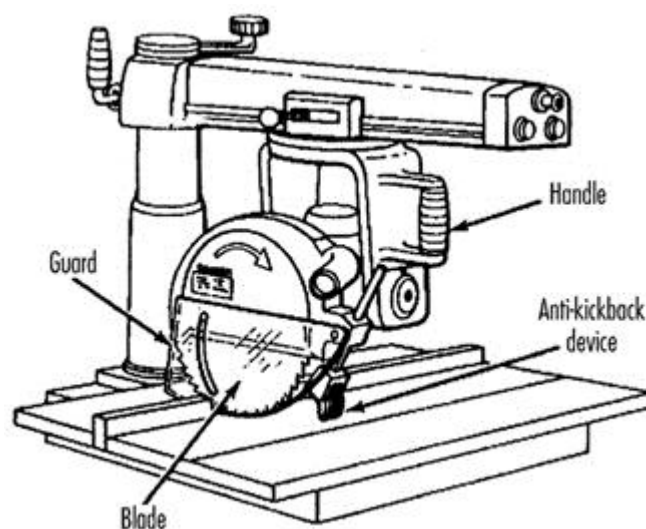


Figure 6. Self-adjusting guard on radial-arm saw

5.2 Safeguarding with devices

Safety devices may stop the machine if a hand or any part of the body is inadvertently placed in the danger area, may restrain or withdraw the operator's hands from the danger area during operation, may require the operator to use both hands on machine controls simultaneously (thus keeping both hands and body out of danger) or may provide a barrier which is synchronized with the operating cycle of the machine in order to prevent entry to the danger area during the hazardous part of the cycle. There are five basic types of safety devices, as follows:

5.3 Presence-sensing devices

Three types of sensing devices which stop the machine or interrupt the work cycle or operation if a worker is within the danger zone are described below:

The *photoelectric (optical) presence-sensing device* uses a system of light sources and controls which can interrupt the machine's operating cycle. If the light field is broken, the machine stops and will not cycle. This device should be used only on machines which can be stopped before the worker reaches the danger area. Figure 7 shows a photoelectric presence-sensing device used with a press brake. The device may be swung up or down to accommodate different production requirements.

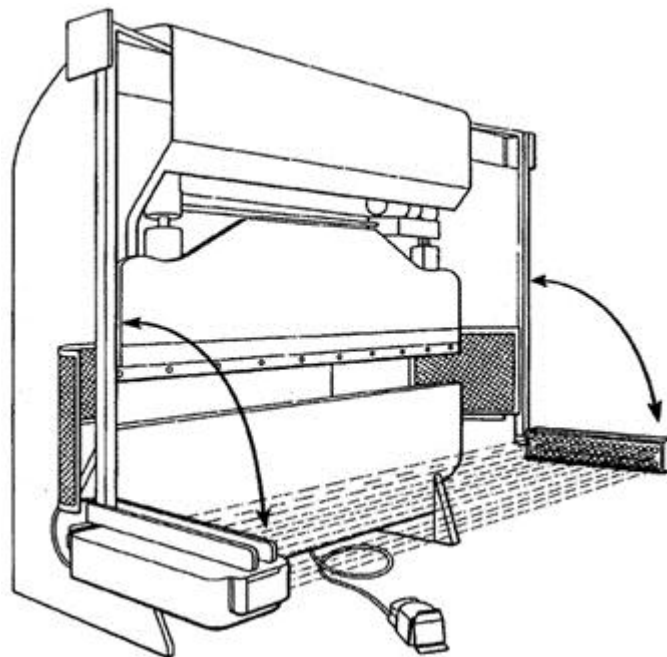


Figure 7. Radio-frequency presence-sensing device on power saw

The *radio-frequency (capacitance) presence-sensing device* uses a radio beam that is part of the control circuit. When the capacitance field is broken, the machine will stop or will not activate. This device should be used only on machines which can be stopped before the worker can reach the danger area. This requires the machine to have a friction clutch or other reliable means for stopping. Figure 8 shows a radio-frequency presence-sensing device mounted on a part-revolution power press.

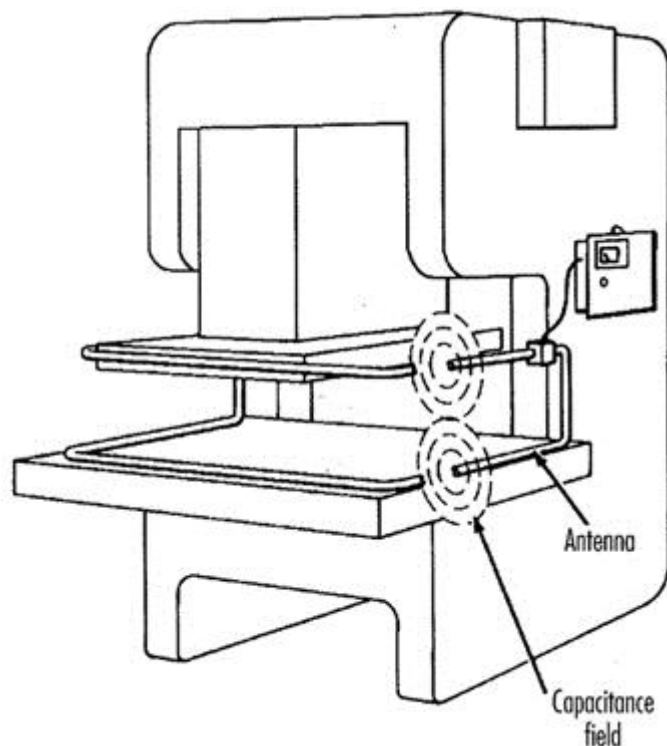


Figure 8 Radio-frequency presence-sensing device mounted on a part-revolution power press.

The *electro-mechanical sensing device* has a probe or contact bar which descends to a predetermined distance when the operator initiates the machine cycle. If there is an obstruction preventing it from descending its full predetermined distance, the control circuit does not actuate the machine cycle. Figure 9 shows an electro-mechanical sensing device on an eyeletter. The sensing probe in contact with the operator's finger is also shown.

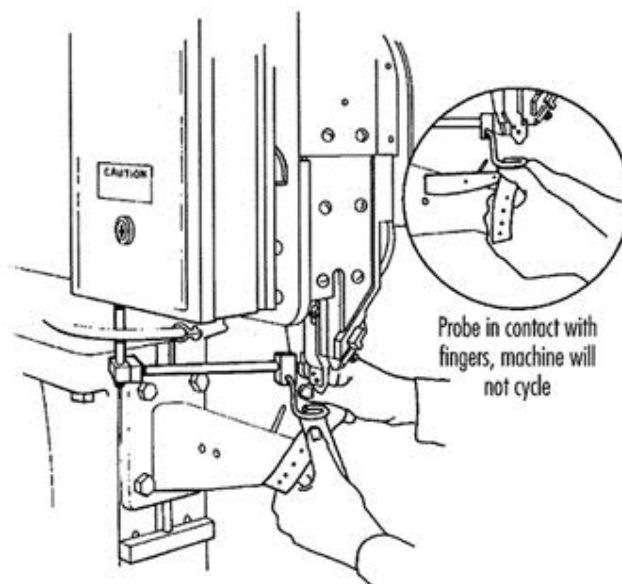


Figure 9. Electromechanical sensing device on eye-letter machine

5.3 Pullback devices

Pullback devices utilize a series of cables attached to the operator's hands, wrists and/or arms and are primarily used on machines with stroking action. When the slide/ram is up, the operator is allowed access to the point of operation. When the slide/ram begins to descend, a mechanical linkage automatically assures withdrawal of the hands from the point of operation. Figure 10 shows a pullback device on a small press.

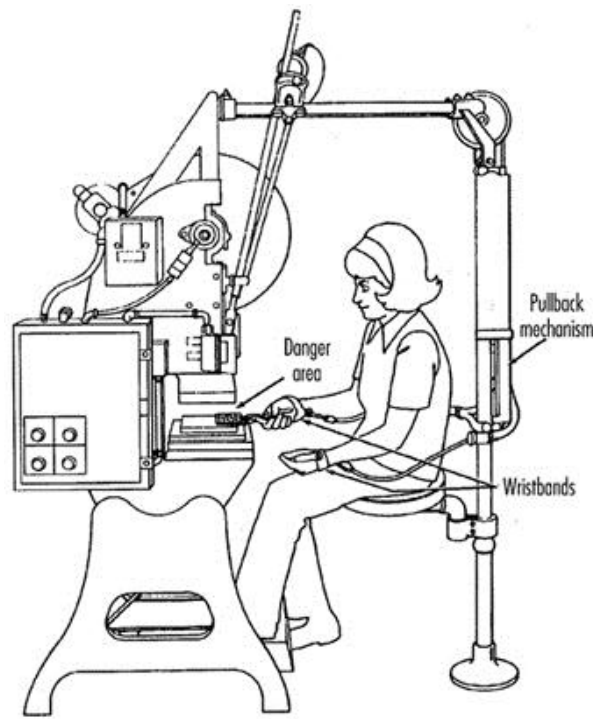


Figure 10. Pullback device on power press

Method	Safeguarding action	Advantages	Limitations
Photoelectric (optical)	· Machine will not start cycling when the light field is interrupted · When the light field is broken by any part of the operator's body during the cycling process, immediate machine braking is activated	Can allow freer movement for operator	· Does not protect against mechanical failure · May require frequent alignment and calibration · Excessive vibration may cause lamp filament damage and premature burnout · Limited to machines that can be stopped without completing cycle
Radio frequency (capacitance)	· Machine cycling will not start when the capacitance field is interrupted · When the capacitance field is disturbed by any part of the operator's body during the cycling process, immediate machine braking is activated	Can allow freer movement for operator	· Does not protect against mechanical failure · Antenna sensitivity must be properly adjusted · Limited to machines that can be stopped without completing cycle
Electro-mechanical	· Contact bar or probe travels a predetermined distance between the operator and the danger area · Interruption of this movement prevents the starting of machine cycle	Can allow access at the point of operation	· Contact bar or probe must be properly adjusted for each application; this adjustment must be maintained properly
Pullback	· As the machine begins to cycle, the operator's hands are pulled out of the danger area	· Eliminates the need for auxiliary barriers or other interference at the danger area	· Limits movement of operator · May obstruct workspace around operator · Adjustments must be made for specific operations and for each individual · Requires frequent inspections and regular maintenance · Requires close supervision of the operator's use of the equipment

Method	Safeguarding action	Advantages	Limitations
Safety trip controls: · Pressure-sensitive body bar · Safety trip-rod · Safety tripwire	· Stops machine when tripped	· Simplicity of use	· All controls must be manually activated · May be difficult to activate controls because of their location · Protects only the operator · May require special fixtures to hold work · May require a machine brake
Two-hand control	· Concurrent use of both hands is required, preventing the operator from entering the danger area	· Operator's hands are at a predetermined location away from danger area · Operator's hands are free to pick up a new part after first half of cycle is completed	· Requires a partial cycle machine with a brake · Some two-hand controls can be rendered unsafe by holding with arm or blocking, thereby permitting one-hand operation · Protects only the operator
Two-hand trip	· Concurrent use of two hands on separate controls prevent hands from being in danger area when machine cycle starts	· Operator's hands are away from danger area · Can be adapted to multiple operations · No obstruction to hand feeding · Does not require adjustment for each operation	· Operator may try to reach into danger area after tripping machine · Some trips can be rendered unsafe by holding with arm or blocking, thereby permitting one-hand operation · Protects only the operator · May require special fixtures
Gate	· Provides a barrier between danger area and operator or other personnel	· Can prevent reaching into or walking into the danger area	· May require frequent inspection and regular maintenance · May interfere with operator's ability to see the work

Table 2. Devices

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SOALAN/QUESTION :

1. What requirement of safeguards?

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2. Why we need to safeguard any machinery required guarding?

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3. Explain three (3) general types of guards?

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